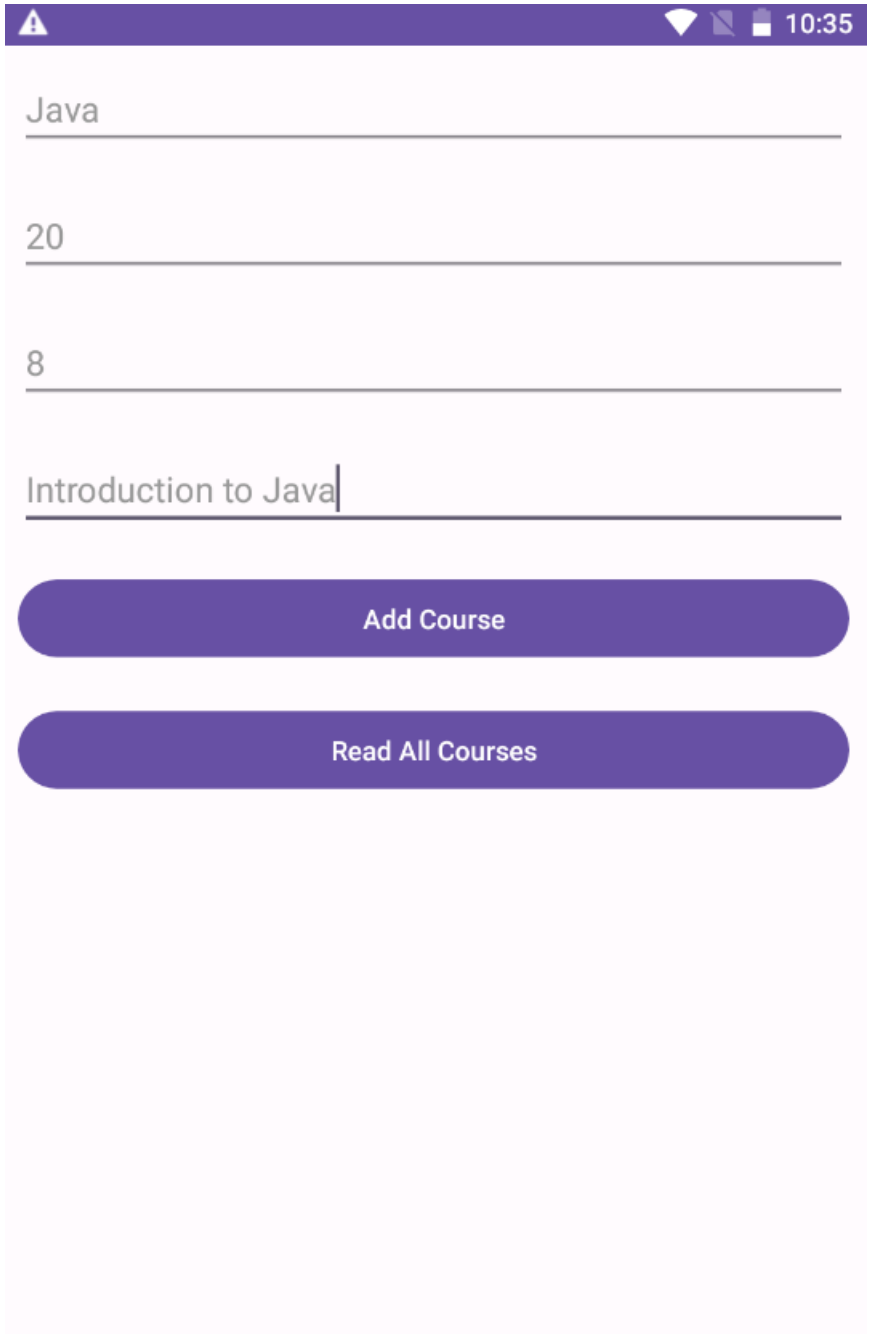
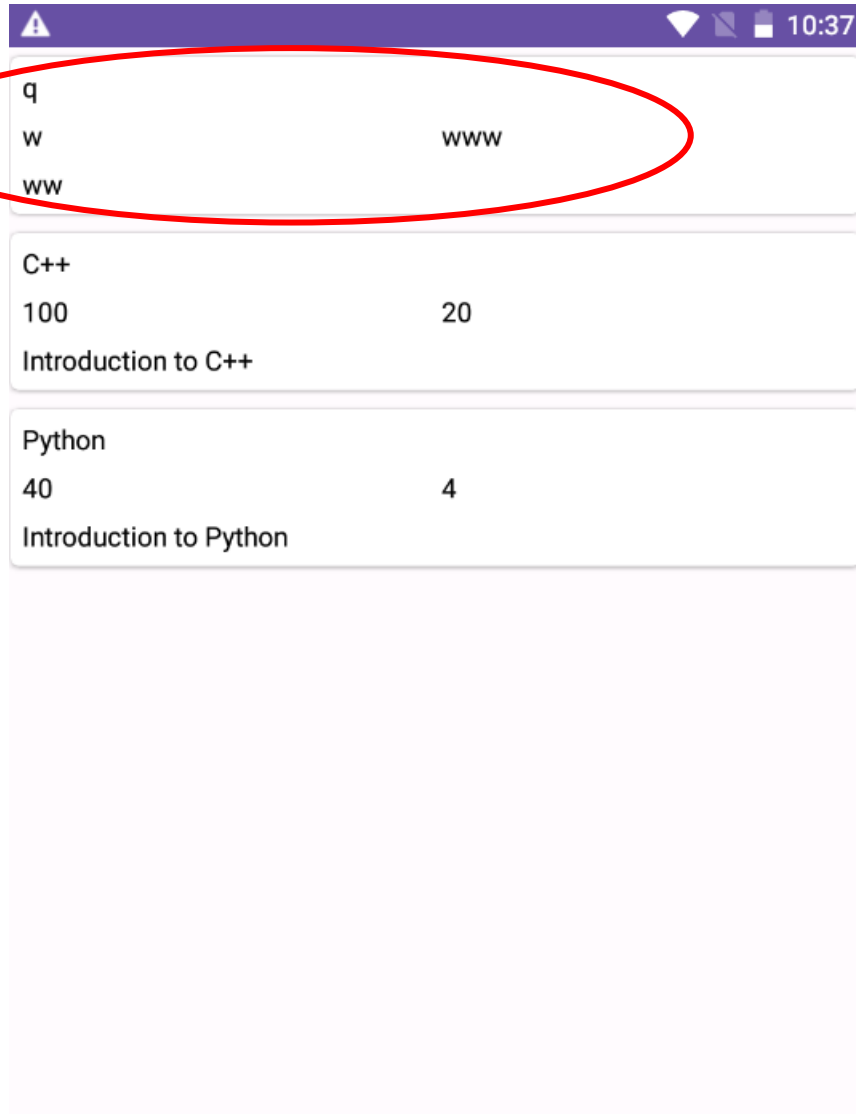
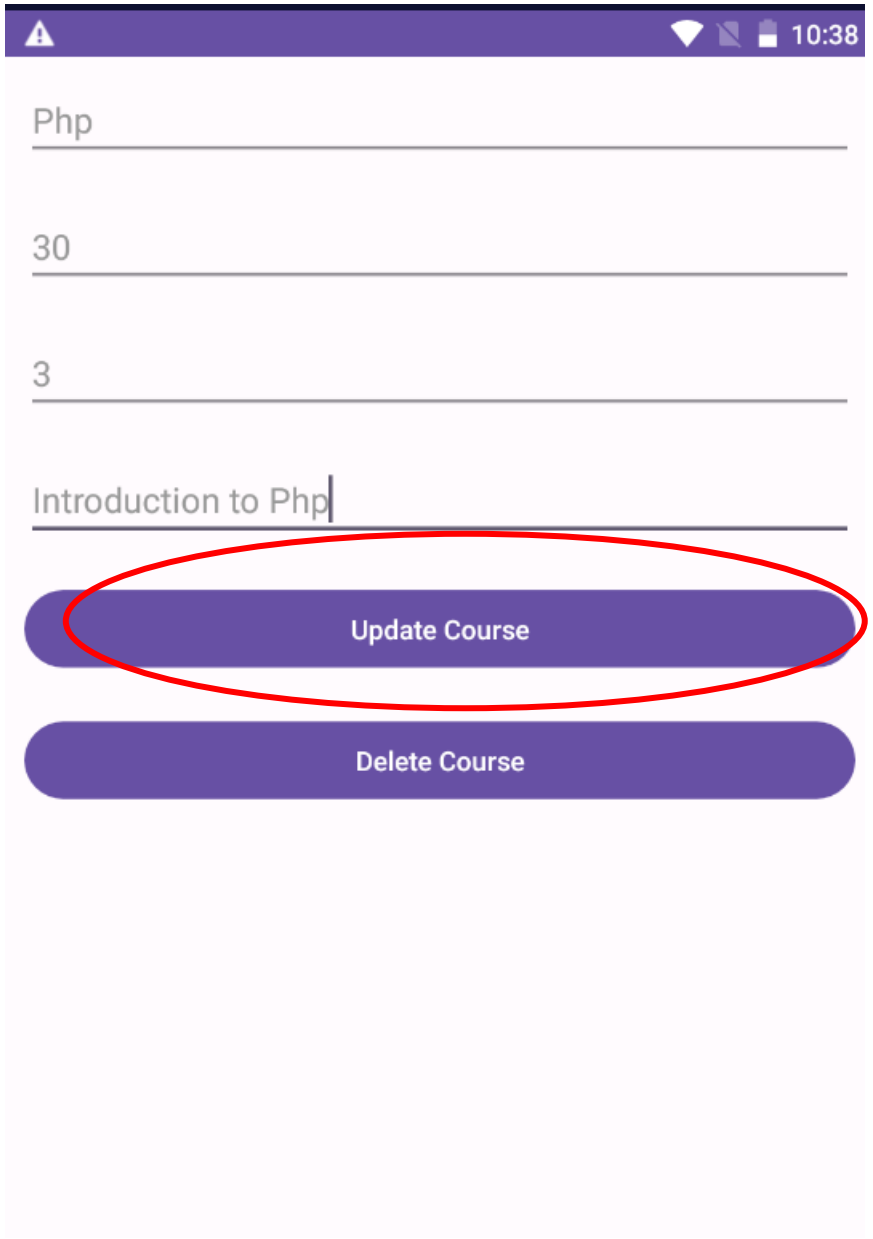
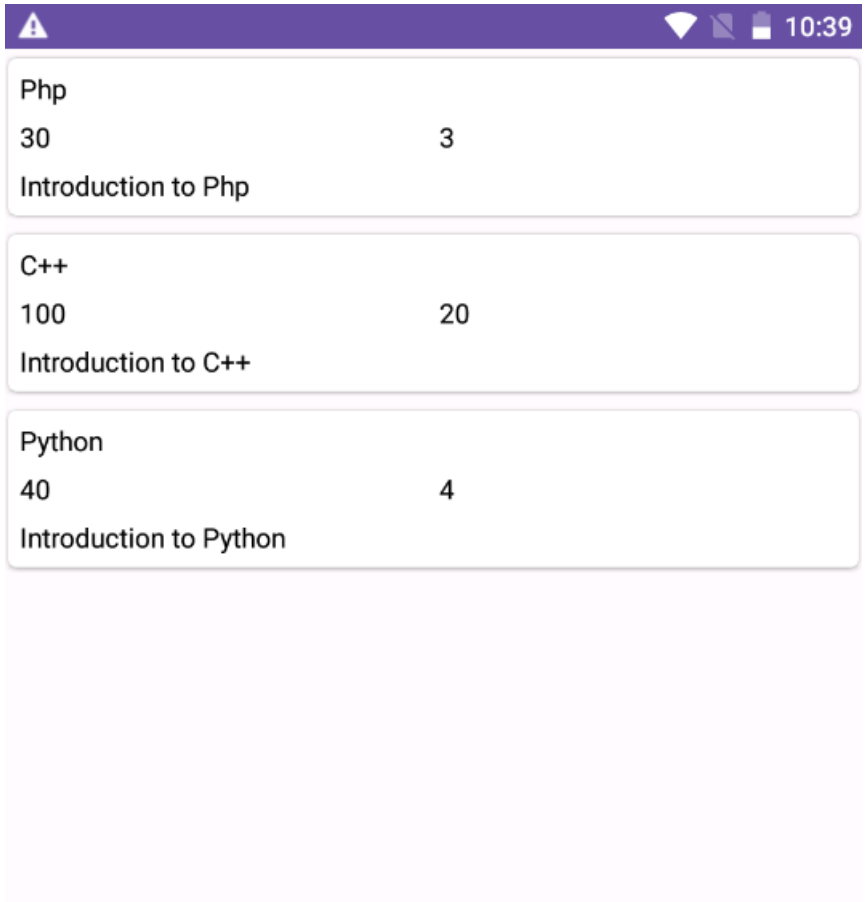
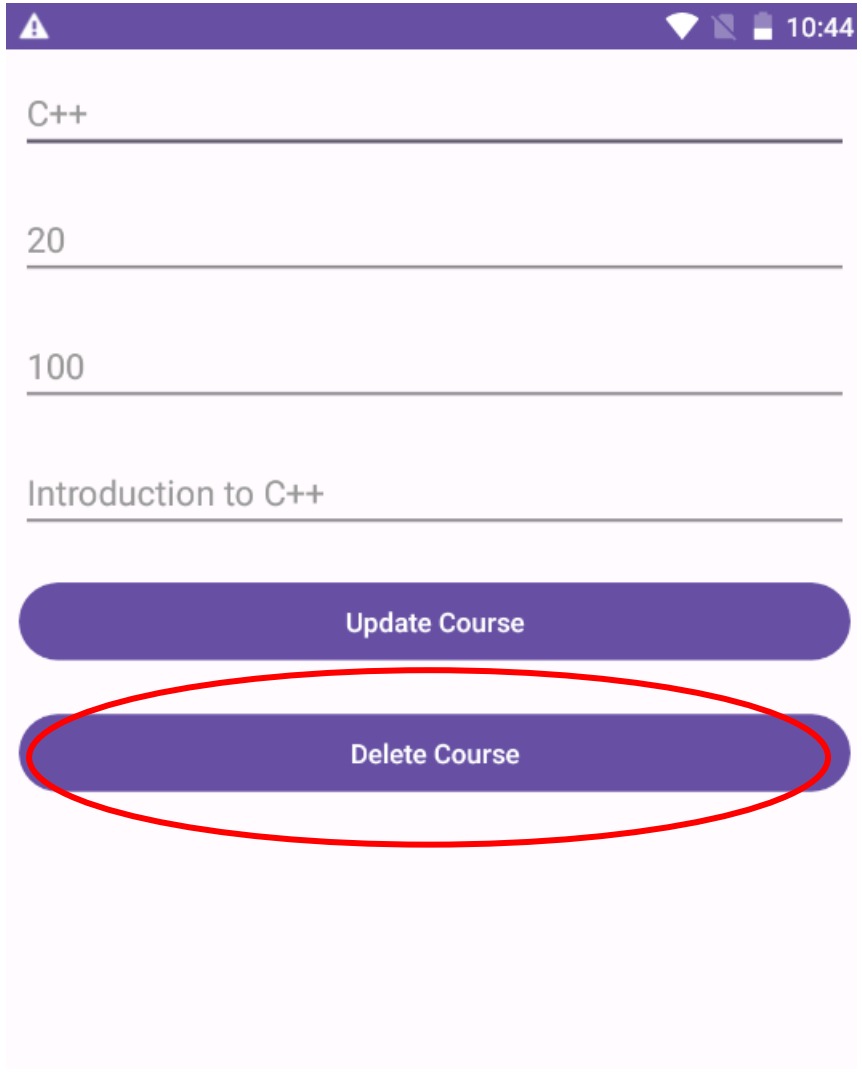
# SQLite Database in Android

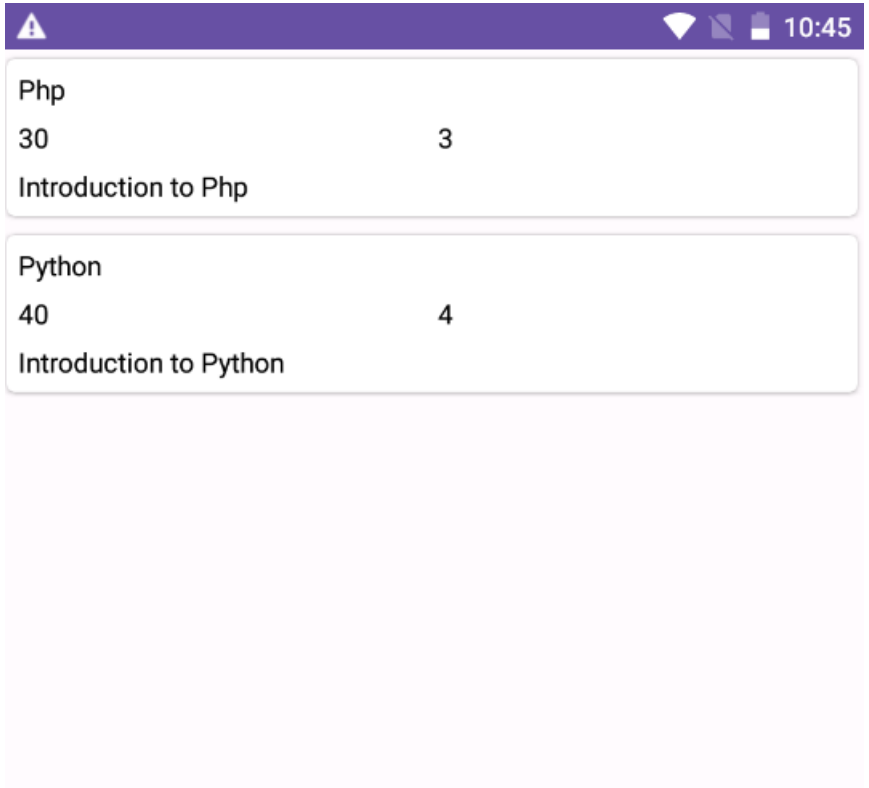

10:35
Java
20
8
Introduction to Java
Add Course
Read All Courses


10:37
q
w
www
ww
C++
100
20
Introduction to C++
Python
40
4
Introduction to Python


10:38
Php
30
3
Introduction to Php
Update Course
Delete Course


10:39
Php
30
3
Introduction to Php
C++
100
20
Introduction to C++
Python
40
4
Introduction to Python


10:44
C++
20
100
Introduction to C++
Update Course
Delete Course


10:45
Php
30
3
Introduction to Php
Python
40
4
Introduction to Python

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:orientation="vertical"
tools:context=".MainActivity">

<EditText
    android:id="@+id/idEdtCourseName"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="10dp"
    android:hint="Enter course Name" />

<EditText
    android:id="@+id/idEdtCourseDuration"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="10dp"
    android:hint="Enter Course Duration" />

<EditText
    android:id="@+id/idEdtCourseTracks"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="10dp"
    android:hint="Enter Course Tracks" />
```

activity_main.xml

```xml
<EditText
    android:id="@+id/idEdtCourseDescription"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="10dp"
    android:hint="Enter Course Description" />

<Button
    android:id="@+id/idBtnAddCourse"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="10dp"
    android:text="Add Course"
    android:textAllCaps="false" />

<Button
    android:id="@+id/idBtnReadCourse"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="10dp"
    android:text="Read All Courses"
    android:textAllCaps="false" />


</LinearLayout>
```

CourseModal.java

```java
package com.example.sqlite2;
//Creating a modal class for storing the data
7 usages
public class CourseModal {

    3 usages
    private String courseName;
    3 usages
    private String courseDuration;
    3 usages
    private String courseTracks;
    3 usages
    private String courseDescription;
    2 usages
    private int id;

    //Getter: To retrieve the value of an object's attribute
    2 usages
    public String getCourseName() { return courseName; }

    2 usages
    public String getCourseDuration() { return courseDuration; }

    2 usages
    public String getCourseTracks() { return courseTracks; }

    2 usages
    public String getCourseDescription() { return courseDescription; }

    public int getId() { return id; }
```

CourseModal.java

```java
    //Setter: To set the value of an object's attribute
    no usages
    public void setCourseName(String courseName) { this.courseName = courseName; }

    no usages
    public void setCourseDuration(String courseDuration) { this.courseDuration = courseDuration; }

    no usages
    public void setCourseTracks(String courseTracks) { this.courseTracks = courseTracks; }

    no usages
    public void setCourseDescription(String courseDescription) {
        this.courseDescription = courseDescription;
    }

    public void setId(int id) { this.id = id; }

    //Constructor: To initialize the values at the time of object creation
    1 usage
    public CourseModal(String courseName, String courseDuration, String courseTracks, String courseDescription) {
        this.courseName = courseName;
        this.courseDuration = courseDuration;
        this.courseTracks = courseTracks;
        this.courseDescription = courseDescription;

    }
}//CourseModal
```

DBHandler.java

```java
package com.example.sqlite2;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import java.util.ArrayList;

6 usages
public class DBHandler extends SQLiteOpenHelper {

    // creating a constant variables for our database.
    1 usage
    private static final String DB_NAME = "coursedb";
    1 usage
    private static final int DB_VERSION = 1;
    6 usages
    private static final String TABLE_NAME = "mycourses";
    1 usage
    private static final String ID_COL = "id";
    3 usages
    private static final String NAME_COL = "name";
    3 usages
    private static final String DURATION_COL = "duration";
    3 usages
    private static final String DESCRIPTION_COL = "description";
    3 usages
    private static final String TRACKS_COL = "tracks";
```

DBHandler.java

```
22      // Constructor for the database handler.
        3 usages
23      public DBHandler(Context context) {
24
25          super(context, DB_NAME,  factory: null, DB_VERSION);
26      }//DBHandler
27
28      // ..............................Creating a database.............
29      @Override
30      public void onCreate(SQLiteDatabase db) {
31          String query = "CREATE TABLE " + TABLE_NAME + " ("
32                  + ID_COL + " INTEGER PRIMARY KEY AUTOINCREMENT, "
33                  + NAME_COL + " TEXT,"
34                  + DURATION_COL + " TEXT,"
35                  + DESCRIPTION_COL + " TEXT,"
36                  + TRACKS_COL + " TEXT)";
37
38          db.execSQL(query);
39      }//onCreate
```

DBHandler.java

```
41      // ..............................Adding a new course..............................
        1 usage
42      public void addNewCourse(String courseName, String courseDuration, String courseDescription, String courseTracks) {
43
44          SQLiteDatabase db = this.getWritableDatabase();
45
46          // creating a variable for content values.
47          ContentValues values = new ContentValues();
48
49          // passing all values along with its key and value pair.
50          values.put(NAME_COL, courseName);
51          values.put(DURATION_COL, courseDuration);
52          values.put(DESCRIPTION_COL, courseDescription);
53          values.put(TRACKS_COL, courseTracks);
54
55          // passing content values to the table.
56          db.insert(TABLE_NAME,  nullColumnHack: null, values);
57
58          // at last we are closing our database after adding database.
59          db.close();
60      }//addNewCourse
```

DBHandler.java

```
62      // ..............................Reading all the courses..............................
        1 usage
63      public ArrayList<CourseModal> readCourses() {
64          // creating a database for reading the database.
65          SQLiteDatabase db = this.getReadableDatabase();
66
67          // creating a cursor with query to read data from database.
68          Cursor cursorCourses = db.rawQuery( sql: "SELECT * FROM " + TABLE_NAME,  selectionArgs: null);
69
70          // creating a new array list.
71          ArrayList<CourseModal> courseModalArrayList = new ArrayList<>();
72
73          // moving the cursor to first position.
74          if (cursorCourses.moveToFirst()) {
75              do {
76                  // adding the data from cursor to the array list.
77                  courseModalArrayList.add(new CourseModal(cursorCourses.getString( columnIndex: 1),
78                          cursorCourses.getString( columnIndex: 4),
79                          cursorCourses.getString( columnIndex: 2),
80                          cursorCourses.getString( columnIndex: 3)));
81              } while (cursorCourses.moveToNext());
82              // moving our cursor to next.
83          }//if
84          // at last closing our cursor and returning our array list.
85          cursorCourses.close();
86          return courseModalArrayList;
87      }//ArrayList
```

DBHandler.java

```
89
90      // ..............................Updating the course..............................
        1 usage
91      public void updateCourse(String originalCourseName, String courseName, String courseDescription,
92                               String courseTracks, String courseDuration) {
93
94          // calling a method to get writable database.
95          SQLiteDatabase db = this.getWritableDatabase();
96          ContentValues values = new ContentValues();
97
98          // passing all values along with its key and value pair.
99          values.put(NAME_COL, courseName);
100         values.put(DURATION_COL, courseDuration);
101         values.put(DESCRIPTION_COL, courseDescription);
102         values.put(TRACKS_COL, courseTracks);
103
104         // calling a update method to update our database and passing our values.
105         // and we are comparing it with name of our course which is stored in original name variable.
106         db.update(TABLE_NAME, values,  whereClause: "name=?", new String[]{originalCourseName});
107         db.close();
108     }
```

```
110         // ..............................Deleting the course..............................
            1 usage
111         public void deleteCourse(String courseName) {
112
113             // Creating a variable to write our database.
114             SQLiteDatabase db = this.getWritableDatabase();
115
116             // Calling a method to delete our course and we are comparing it with our course name.
117             db.delete(TABLE_NAME,  whereClause: "name=?", new String[]{courseName});
118             db.close();
119         }
120
            10 usages
121         @Override
122         public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
123             // this method is called to check if the table exists already.
124             db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
125             onCreate(db);
126         }//onUpgrade
127
128 }//DBHandler
```

MainActivity.java

```
3   import android.os.Bundle;
4   import android.content.Intent;
5   import android.view.View;
6   import android.widget.Button;
7   import android.widget.EditText;
8   import android.widget.Toast;
9   import androidx.appcompat.app.AppCompatActivity;
10
11  //Creating an Activity for displaying the list of courses
12  public class MainActivity extends AppCompatActivity {
13
        3 usages
14      private EditText courseNameEdt, courseTracksEdt, courseDurationEdt, courseDescriptionEdt;
        2 usages
15      private Button addCourseBtn, readCourseBtn,updateCourseBtn, deleteCourseBtn;
        2 usages
16      private DBHandler dbHandler;
        no usages
17      String courseName, courseDesc, courseDuration, courseTracks;
```

MainActivity.java

```
20      @Override
21      protected void onCreate(Bundle savedInstanceState) {
22          super.onCreate(savedInstanceState);
23          setContentView(R.layout.activity_main);
24
25          courseNameEdt = findViewById(R.id.idEdtCourseName);
26          courseTracksEdt = findViewById(R.id.idEdtCourseTracks);
27          courseDurationEdt = findViewById(R.id.idEdtCourseDuration);
28          courseDescriptionEdt = findViewById(R.id.idEdtCourseDescription);
29          addCourseBtn = findViewById(R.id.idBtnAddCourse);
30          readCourseBtn = findViewById(R.id.idBtnReadCourse);
31
32          // creating a new dbhandler class and passing the context to it.
33          dbHandler = new DBHandler( context: MainActivity.this);
34
35          addCourseBtn.setOnClickListener(new View.OnClickListener() {
36              @Override
37              public void onClick(View v) {
38
39                  String courseName = courseNameEdt.getText().toString();
40                  String courseTracks = courseTracksEdt.getText().toString();
41                  String courseDuration = courseDurationEdt.getText().toString();
42                  String courseDescription = courseDescriptionEdt.getText().toString();
43
44                  // validating if the text fields are empty or not.
45                  if (courseName.isEmpty() || courseTracks.isEmpty() || courseDuration.isEmpty() || courseDescription.isEmpty()) {
46                      Toast.makeText( context: MainActivity.this,  text: "Please enter all the data..", Toast.LENGTH_SHORT).show();
47                      return;
48                  }//if
```

```
                // calling a method to add new course to sqlite data and pass all our values to it.
                dbHandler.addNewCourse(courseName, courseDuration, courseDescription, courseTracks);

                // after adding the data, a toast message is displayed.
                Toast.makeText( context: MainActivity.this,  text: "Course has been added.", Toast.LENGTH_SHORT).show();
                courseNameEdt.setText("");
                courseDurationEdt.setText("");
                courseTracksEdt.setText("");
                courseDescriptionEdt.setText("");
            }////onClick
        });//setOnClickListener

        readCourseBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                // opening a new activity via a intent.
                Intent i = new Intent( packageContext: MainActivity.this, ViewCourses.class);
                startActivity(i);
            }//onClick
        });//setOnClickListener

    }//onCreate
}//MainActivity
```

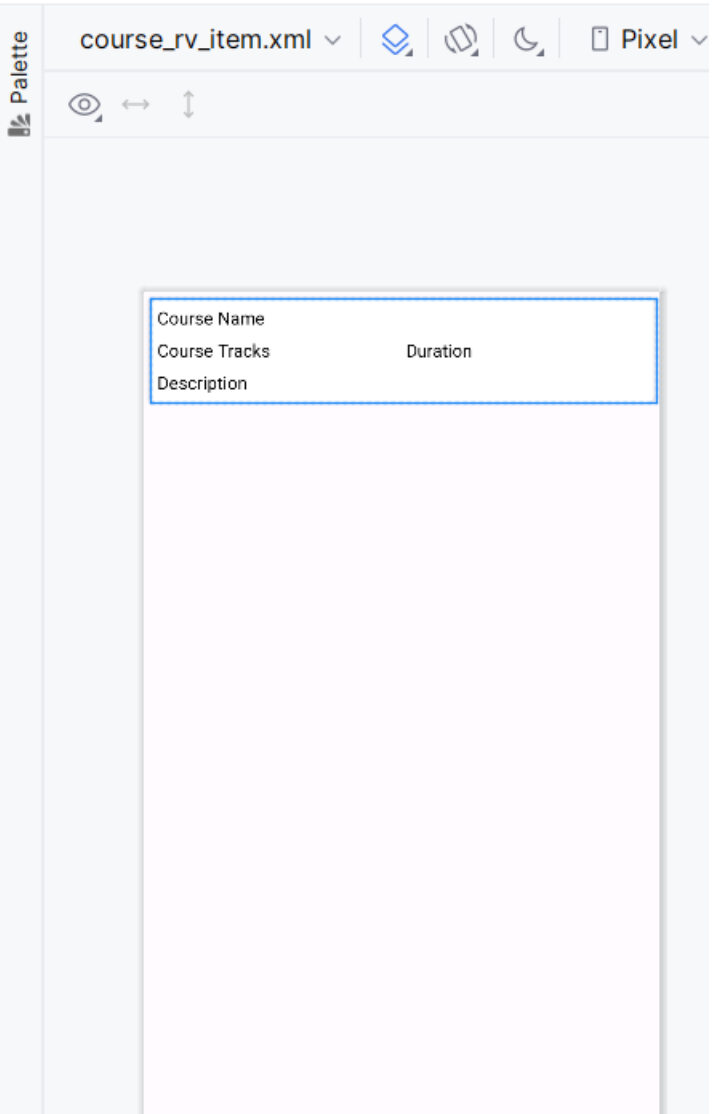
course_rv_item.xml

```xml
<?xml version="1.0" encoding="utf-8"?>

<!--> Creating a new layout file for the item of RecyclerView <!
<androidx.cardview.widget.CardView
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="5dp"
    android:elevation="8dp"
    app:cardCornerRadius="4dp">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="3dp"
        android:orientation="vertical">

        <TextView
            android:id="@+id/idTVCourseName"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:padding="3dp"
            android:text="Course Name"
            android:textColor="@color/black" />
```



course_rv_item.xml

```xml
        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="horizontal"
            android:weightSum="2">

            <TextView
                android:id="@+id/idTVCourseTracks"
                android:layout_width="0dp"
                android:layout_height="wrap_content"
                android:layout_weight="1"
                android:padding="3dp"
                android:text="Course Tracks"
                android:textColor="@color/black" />

            <TextView
                android:id="@+id/idTVCourseDuration"
                android:layout_width="0dp"
                android:layout_height="wrap_content"
                android:layout_weight="1"
                android:padding="3dp"
                android:text="Duration"
                android:textColor="@color/black" />

        </LinearLayout>

        <TextView
            android:id="@+id/idTVCourseDescription"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:padding="3dp"
            android:text="Description"
            android:textColor="@color/black" />

    </LinearLayout>

</androidx.cardview.widget.CardView>
```

CourseRVAdapter.java

```java
import android.content.Context;
import android.content.Intent;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;
import java.util.ArrayList;

3 usages
public class CourseRVAdapter extends RecyclerView.Adapter<CourseRVAdapter.ViewHolder> {

    3 usages
    private ArrayList<CourseModal> courseModalArrayList;
    3 usages
    private Context context;

    // constructor
    1 usage
    public CourseRVAdapter(ArrayList<CourseModal> courseModalArrayList, Context context) {
        this.courseModalArrayList = courseModalArrayList;
        this.context = context;
    }

    @NonNull
    @Override
    public ViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int viewType) {
        // Inflating the layout file for the recycler view items.
        View view = LayoutInflater.from(parent.getContext()).inflate(R.layout.course_rv_item, parent, attachToRoot: false);
        return new ViewHolder(view);
    }
```

CourseRVAdapter.java

```java
@Override
public void onBindViewHolder(@NonNull ViewHolder holder, int position) {
    // Setting data to the views of recycler view item.
    CourseModal modal = courseModalArrayList.get(position);
    holder.courseNameTV.setText(modal.getCourseName());
    holder.courseDescTV.setText(modal.getCourseDescription());
    holder.courseDurationTV.setText(modal.getCourseDuration());
    holder.courseTracksTV.setText(modal.getCourseTracks());

    // to add on click listener for the recycler view item.
    holder.itemView.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {

            // Calling an intent.
            Intent i = new Intent(context, UpdateCourseActivity.class);

            // Passing all our values.
            i.putExtra( name: "name", modal.getCourseName());
            i.putExtra( name: "description", modal.getCourseDescription());
            i.putExtra( name: "duration", modal.getCourseDuration());
            i.putExtra( name: "tracks", modal.getCourseTracks());

            context.startActivity(i);
        }
    });
}
```

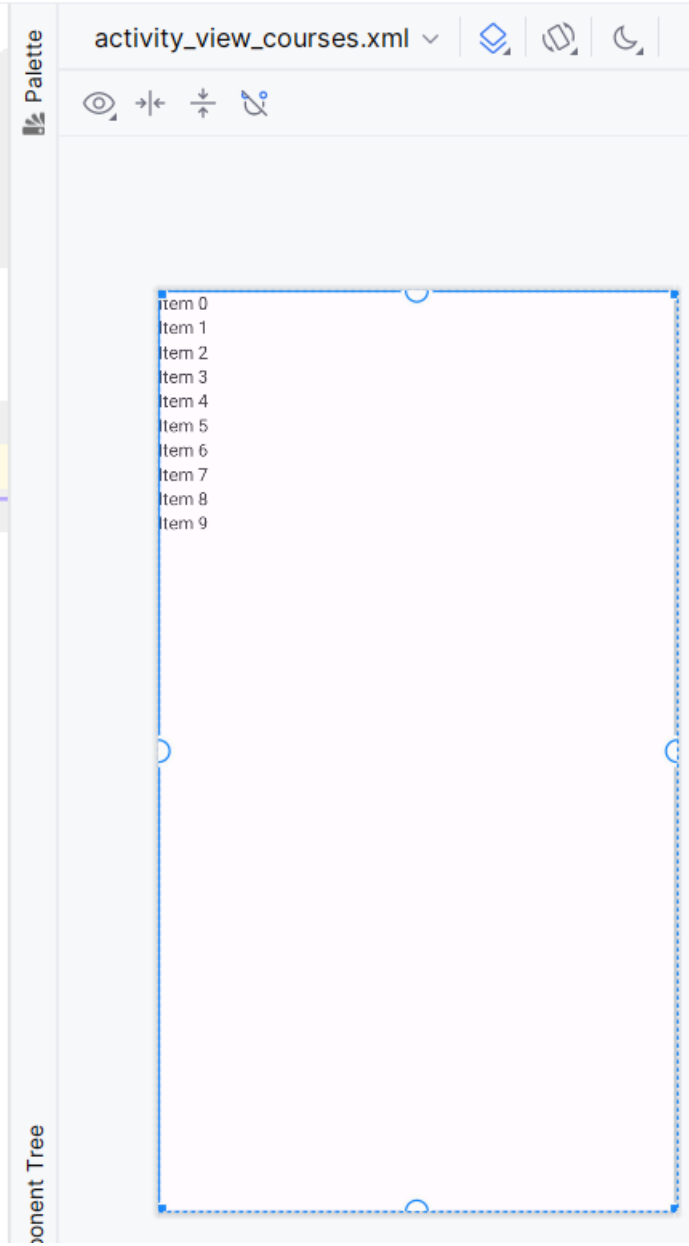
```
60      @Override
61      public int getItemCount() {
62          // returning the size of our array list
63          return courseModalArrayList.size();
64      }
65
        4 usages
66      public class ViewHolder extends RecyclerView.ViewHolder {
67
68          // creating variables for our text views.
            2 usages
69          private TextView courseNameTV, courseDescTV, courseDurationTV, courseTracksTV;
70
            1 usage
71          public ViewHolder(@NonNull View itemView) {
72              super(itemView);
73              // initializing our text views
74              courseNameTV = itemView.findViewById(R.id.idTVCourseName);
75              courseDescTV = itemView.findViewById(R.id.idTVCourseDescription);
76              courseDurationTV = itemView.findViewById(R.id.idTVCourseDuration);
77              courseTracksTV = itemView.findViewById(R.id.idTVCourseTracks);
78          }
79      }//ViewHolder
80  }//CourseRVAdapter
```

activity_view_courses.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".ViewCourses">

    <!--recycler view for displaying our courses-->
    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/idRVCourses"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</RelativeLayout>
```



ViewCourses.java

```java
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;
import java.util.ArrayList;

public class ViewCourses extends AppCompatActivity {

    // creating variables for the array list,dbhandler, adapter and recycler view.
    3 usages
    private ArrayList<CourseModal> courseModalArrayList;
    2 usages
    private DBHandler dbHandler;
    2 usages
    private CourseRVAdapter courseRVAdapter;
    3 usages
    private RecyclerView coursesRV;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_view_courses);

        // initializing all variables.
        courseModalArrayList = new ArrayList<>();
        dbHandler = new DBHandler( context: ViewCourses.this);

        // getting our course array list from dbhandler class.
        courseModalArrayList = dbHandler.readCourses();
```

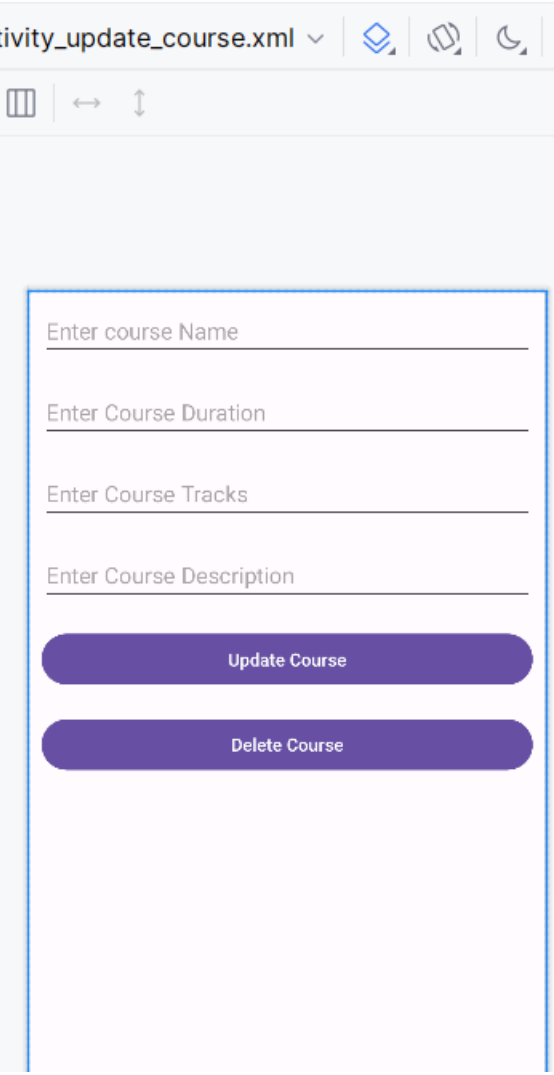
```
28
29          // passing the array list to the adapter class.
30          courseRVAdapter = new CourseRVAdapter(courseModalArrayList,  context: ViewCourses.this);
31          coursesRV = findViewById(R.id.idRVCourses);
32
33          // setting layout manager for the recycler view.
34          LinearLayoutManager linearLayoutManager = new LinearLayoutManager( context: ViewCourses.this, RecyclerView.VERTICAL,  reverseLayout: false);
35          coursesRV.setLayoutManager(linearLayoutManager);
36
37          // setting the adapter to recycler view.
38          coursesRV.setAdapter(courseRVAdapter);
39      }//onCreate
40  }//ViewCourses
```

activity_update_course.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".UpdateCourseActivity">

    <!--Edit text to enter course name-->
    <EditText
        android:id="@+id/idEdtCourseName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:hint="Enter course Name" />

    <!--edit text to enter course duration-->
    <EditText
        android:id="@+id/idEdtCourseDuration"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:hint="Enter Course Duration" />
```



activity_update_course.xml

```
    <!--edit text to display course tracks-->
    <EditText
        android:id="@+id/idEdtCourseTracks"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:hint="Enter Course Tracks" />

    <!--edit text for course description-->
    <EditText
        android:id="@+id/idEdtCourseDescription"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:hint="Enter Course Description" />

    <!--button for updating our course-->
    <Button
        android:id="@+id/idBtnUpdateCourse"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:text="Update Course"
        android:textAllCaps="false" />

    <!--button for deleting our course-->
    <Button
        android:id="@+id/idBtnDelete"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="10dp"
        android:text="Delete Course"
        android:textAllCaps="false" />

</LinearLayout>
```

UpdateCourseActivity.java

```java
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class UpdateCourseActivity extends AppCompatActivity {

    3 usages
    private EditText courseNameEdt, courseTracksEdt, courseDurationEdt, courseDescriptionEdt;
    2 usages
    private Button updateCourseBtn, deleteCourseBtn;
    3 usages
    private DBHandler dbHandler;
    4 usages
    String courseName, courseDesc, courseDuration, courseTracks;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_update_course);

        courseNameEdt = findViewById(R.id.idEdtCourseName);
        courseTracksEdt = findViewById(R.id.idEdtCourseTracks);
        courseDurationEdt = findViewById(R.id.idEdtCourseDuration);
        courseDescriptionEdt = findViewById(R.id.idEdtCourseDescription);
        updateCourseBtn = findViewById(R.id.idBtnUpdateCourse);
        deleteCourseBtn = findViewById(R.id.idBtnDelete);
```

UpdateCourseActivity.java

```
30        // Initializing our dbhandler class.
31        dbHandler = new DBHandler( context: UpdateCourseActivity.this);
32
33        // Getting data
34        courseName = getIntent().getStringExtra( name: "name");
35        courseDesc = getIntent().getStringExtra( name: "description");
36        courseDuration = getIntent().getStringExtra( name: "duration");
37        courseTracks = getIntent().getStringExtra( name: "tracks");
38
39        // setting data
40        courseNameEdt.setText(courseName);
41        courseDescriptionEdt.setText(courseDesc);
42        courseTracksEdt.setText(courseTracks);
43        courseDurationEdt.setText(courseDuration);
44
45        // adding on click listener to our update course button.
46        updateCourseBtn.setOnClickListener(new View.OnClickListener() {
47            @Override
48            public void onClick(View v) {
49
50                // Calling an update course method and passing all the edit text values.
51                dbHandler.updateCourse(courseName, courseNameEdt.getText().toString(), courseDescriptionEdt.getText().toString(),
52                        courseTracksEdt.getText().toString(), courseDurationEdt.getText().toString());
53
54                Toast.makeText( context: UpdateCourseActivity.this,  text: "Course Updated..", Toast.LENGTH_SHORT).show();
55
56                // launching our main activity.
57                Intent i = new Intent( packageContext: UpdateCourseActivity.this, MainActivity.class);
58                startActivity(i);
59            }//onClick
60        });//setOnClickListener
```

```
62          // adding on click listener for delete button to delete our course.
63          deleteCourseBtn.setOnClickListener(new View.OnClickListener() {
64              @Override
65              public void onClick(View v) {
66                  // calling a method to delete the course.
67                  dbHandler.deleteCourse(courseName);
68                  Toast.makeText( context: UpdateCourseActivity.this,  text: "Deleted the course", Toast.LENGTH_SHORT).show();
69                  Intent i = new Intent( packageContext: UpdateCourseActivity.this, MainActivity.class);
70                  startActivity(i);
71              }//onClick
72          });//setOnClickListener
73      }//onCreate
74  }//UpdateCourseActivity
```